# TITLE OF YOUR PROJECT

A PROJECT REPORT

Submitted in partial fulfillment for the Degree of

Bachelor of Technology under the

School of Computing

Submitted by

| Register No | Names of Students |
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# BONAFIDE CERTIFICATE

This is to certify that this is a bonafide record of the project presented by the students whose names are given below during Project Phase I (22AIE498) in partial fulfilment of the requirements of the degree of Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence).

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# Acknowledgements

Write your acknowledgments in this page.

(your name in CAPS)

# List of Figures

# List of Tables

# Abstract

Write your abstract here

# Chapter 1

# Introduction

## 1.1 Background

Provide a broad overview of the project domain, its importance, and context. Introduce the main theme of the research/project, including key terms and general trends in the field.

## 1.2 Problem Addressed

Clearly define the problem or challenge the project aims to address. Discuss the gap, limitation, or need in the current system, research, or practice. Tip: Use precise and concise language to articulate the issue.

## 1.3 Motivation

Outline the goals the project aims to achieve. Include main objectives (broad goals). If applicable, list specific objectives (detailed steps or milestones).

## 1.4 Scope of the Study/Project

Define the boundaries and focus of the project. Specify what is included and excluded. Highlight the target domain, dataset, or systems under consideration.

Conclude by discussing the organization of the project report. Briefly describe what each chapter covers. Ensure the flow is logical and coherent.

# Chapter 2

# Literature Review/ Existing System

## 2.1 Existing System Study

### 2.1.1 Review of existing Research work

The review of existing research work provides an overview of past studies and developments relevant to the project topic, highlighting key findings, methodologies, and research gaps.

### 2.1.2 Review of Existing Tools/Products

The review of existing tools/products analyzes current solutions available in the market or research domain that address similar problems. This helps in understanding their features, limitations, and how the proposed project can offer improvements or unique contributions.

Review existing products, tools, or technologies solving similar problems. Include market comparisons or feature analysis.

## 2.2 Research Gap/ Scope for improvement and innovation

Limitations/ problems not addressed in existing systems Point out the specific shortcomings or challenges that persist in existing solutions.Identify areas or research questions that have not been addressed

1. Which are all the gaps that you have addressed now in the Phase I
2. Gaps which you plan to address in phase II( If the project is to be continued)

## 2.3 Problem Statement and Contributions

1. Problem Statement: Clearly articulate the specific challenge you aim to solve. Ensure it flows naturally from the gaps identified in the previous slide. Use precise, concise

language to define the problem scope.

2. Research Contributions Present 2–4 clear, measurable research contribution of your work. For Research Projects: Focus on advancing knowledge, developing methodologies, or validating hypotheses. For Product-Based Projects: Focus on delivering a functional, innovative solution with specific features or improvements.

# Chapter 3

# Proposed Work

## 3.1 Proposed Work

Provide an overview of the proposed work, connecting it to the research gaps identified in the Literature Study chapter. Briefly restate the problem. Highlight how your proposed work addresses the gaps or challenges. Emphasize the novelty or uniqueness of your approach.

### 3.1.1 Objectives of the Proposed Work

Clearly articulate the specific objectives of the proposed research or system.

Define measurable goals. Include both primary and secondary objectives.

## 3.2 Methodology

Describe the methods, techniques, and tools to be employed. Provide a detailed explanation of your approach. Include algorithms, frameworks, models, and workflows. Use diagrams to explain the methodology clearly.

### 3.2.1 Overview of the Approach:

A summary of your method.

### 3.2.2 Dataset Selection:

Describe the datasets (real-world, synthetic, or benchmarks) used for evaluation.

### 3.2.3 Algorithm/Model Design:

Explain the model architecture, algorithmic steps, or computational processes.

### 3.2.4 Tools and Technologies:

Mention the programming languages, libraries, or software used.System Architecture/Design (Optional, for implementation-based projects) Provide a high-level view of the system's components and interactions. Include block diagrams or flowcharts to illustrate the design. Describe individual components and their roles.

### 3.2.5 Algorithm or Model Description

Explain the model architecture, algorithmic steps, or computational processes.

Also give explanation of how these algorithms are used in solving the problems. The prerequisites, if any, need to be explained. Any equations to clarify the algorithms, if necessary, may also be given.

A sample algorithm is shown in Algorithm 1

**Algorithm 1** Pseudocode for existing model

1: **Input** Subset of Features with corresponding values; Required Target feature name
2: **Output** Target feature value
3: Retrieve the coefficient values from the database for the given input subset of features
4: **for** each record in the features' coefficient values **do**
5: Apply linear regression
6: $h_\theta(x) \leftarrow \theta^T.X$, where $\theta$ and X are column matrices
7: **end for**

Provide detailed steps of the proposed algorithm or architecture. Outline the algorithm step-by-step. Include pseudocode or flowcharts for clarity. Explain any mathematical formulations or equations involved.

### 3.2.6 Expected Outcomes

Discuss the anticipated results and how they address the research objectives. Highlight the expected performance improvements or practical contributions.

### 3.2.7 Advantages of the Proposed Work

Emphasize the benefits of the proposed approach. Compare with existing methods, highlighting improvements. Mention scalability, efficiency, accuracy, or usability gains.

### 3.2.8 Limitations and Assumptions

Acknowledge potential limitations and assumptions made in the study.

Mention constraints like computational resources, data availability, or generalizability. Discuss any assumptions inherent in the model or methodology.

# Chapter 4

# Experimentation and Result Analysis

Describe how the proposed work was evaluated, and provide an analysis of the results.

## 4.1 Experimental Setup

Describe the resources, tools, and procedures used for conducting experiments.

Hardware and Software: Specify the computational resources, programming languages, libraries, or platforms used. Example: "Experiments were conducted on a system with an Intel i7 processor, 16GB RAM, using Python 3.9 and PyTorch."

## 4.2 Datasets:

Provide details about the datasets used, including their source, characteristics (e.g., number of nodes, edges, features), and preprocessing steps. Example: "The experiments used two datasets: the Facebook Social Network dataset and the DBLP Citation Network, both containing time-stamped edges."

## 4.3 Evaluation Metrics:

Define the metrics used to measure performance, such as precision, recall, F1 score, ROC-AUC, or computational efficiency. Example: "Precision and recall were used to evaluate prediction accuracy, while runtime was used to measure computational efficiency."

## 4.4 Experimental Design

Describe how the experiments were conducted step by step.: Baseline Methods: List the existing methods used for comparison. Example: "The proposed model was compared against node2vec, GCN, and GAT models."

### 4.4.1 Experimental Scenarios:

Discuss different conditions under which the experiments were conducted, such as varying dataset sizes, hyperparameters, or network dynamics. Example: "The experiments evaluated the model under static and dynamic conditions by incrementally adding edges over time."

### 4.4.2 Parameter Tuning:

Mention how hyperparameters were optimized, if applicable. Example: "Grid search was performed to tune the learning rate, dropout rate, and number of GNN layers."

## 4.5 Results

Present the results of your experiments in a clear and organized manner.

Tables and Graphs: Use tables, charts, and graphs to present quantitative results. Comparative Results: Compare the performance of your proposed model with baseline methods. Example: "The proposed model achieved an F1 score of 0.89 compared to 0.83 for node2vec." Qualitative Results (if applicable): Include visualizations, case studies, or examples of outputs. Example: "Figure 4.2 shows the network before and after link prediction using the proposed model."

## 4.6 Analysis of Results

Interpret the results and explain their significance.

## 4.7 Observations:

Discuss patterns or anomalies observed in the results. Example: "The proposed model outperforms baseline methods for dense graphs but shows reduced performance for sparse graphs." Insights: Relate the results back to the research objectives and literature gaps. Example: "The results validate the proposed hybrid approach's effectiveness in handling dynamic networks, addressing the scalability issue highlighted in the literature." Statistical Significance (if applicable): Include significance testing to strengthen claims.

## 4.8 Comparative Analysis

Provide a side-by-side comparison of your results with those of existing methods.

Include a table summarizing metrics for all methods. Discuss the relative advantages or limitations of your approach. Example: "While the runtime of the proposed model is slightly higher than node2vec, it achieves significantly better precision."

# Chapter 5

# Conclusion and Scope for further Research

# Appendix A

# title of appendix

Include your Code here

# References

1. Mr. X, Mr. U, Mr. V and Mr.Z , “Efficient Computations in Operating Systems”, (Communicated in Sept. 2018).